

AKQI, China

WMO: 517110

Lat: 40.9414N

Lon: 78.4512E

Elev: 1986

StdP: 79.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-16.3	-23.8	0.6	-7.1	-21.9	0.7	-8.5	5.9	-5.2	5.3	-5.7	1.7	250	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	28.7	14.0	27.3	13.4	25.9	13.0	15.8	24.7	15.1	23.7	14.4	22.8	2.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	12.0	18.5	12.0	11.2	17.9	11.1	10.5	17.5	52.0	24.5	49.5	23.8	47.3	22.7	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.5	5.0	4.4		-19.9	32.2	2.6	1.3	-21.8	33.1	-23.3	33.9	-24.8	34.6	-26.6	35.5	
				-20.3	17.4	2.5	1.5	-22.2	18.5	-23.6	19.3	-25.0	20.2	-26.8	21.2	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-8.4	-4.5	3.6	10.5	14.7	18.0	19.9	19.0	14.9	8.6	1.3	-5.8
	DBStd	10.43	4.40	4.28	4.44	4.21	3.36	2.68	2.63	2.64	3.09	3.53	3.97	4.15
	HDD10.0	2052	571	407	202	43	6	0	0	0	3	69	262	490
	HDD18.3	4022	829	641	457	234	117	37	13	24	109	301	511	748
	CDD10.0	1216	0	0	3	59	151	241	307	278	149	27	0	0
	CDD18.3	145	0	0	0	1	5	28	62	44	5	0	0	0
	CDH23.3	1096	0	0	0	11	49	214	472	307	42	1	0	0
	CDH26.7	193	0	0	0	1	2	28	106	54	3	0	0	0
Wind	WSAvg	2.3	2.2	2.3	2.5	2.6	2.3	2.4	2.2	2.2	2.1	2.1	2.2	2.2
Precipitation	PrecAvg	187	2	4	11	15	26	28	33	38	22	3	2	1
	PrecMax	350	7	22	42	60	67	86	98	103	66	22	14	14
	PrecMin	89	0	0	0	0	0	3	6	0	0	0	0	0
	PrecStd	60	2	5	12	15	17	25	23	28	16	5	4	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	8.7	17.8	24.9	26.4	29.1	31.5	30.2	26.7	21.3	14.0	8.0
		MCWB	-1.2	0.5	5.7	9.5	10.8	13.1	14.8	14.9	12.6	9.2	4.6	0.4
	2%	DB	2.3	5.6	15.1	22.2	24.8	27.3	29.3	28.2	24.4	18.6	11.3	4.2
		MCWB	-3.3	-1.0	4.3	8.5	10.4	12.6	14.3	14.2	11.6	8.1	2.8	-1.8
	5%	DB	-0.2	3.4	12.9	20.2	23.3	25.9	27.7	26.5	22.6	16.7	9.2	1.7
		MCWB	-4.8	-2.0	3.5	7.7	9.9	12.3	13.9	13.8	11.3	7.2	2.1	-3.3
	10%	DB	-2.5	1.6	10.7	17.9	21.6	24.3	26.1	25.0	20.9	14.6	7.1	-0.3
		MCWB	-6.1	-3.0	2.5	6.6	9.5	11.9	13.6	13.5	10.8	6.2	1.0	-4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	0.8	6.2	10.4	12.6	15.4	17.1	16.9	14.1	10.1	4.8	0.5
		MCDB	5.5	7.2	16.2	22.5	22.1	25.3	26.4	25.1	23.0	18.4	12.8	7.2
	2%	WB	-3.1	-0.7	4.9	9.1	11.6	14.1	15.9	15.9	13.0	8.9	3.5	-1.7
		MCDB	2.0	4.8	13.4	20.2	21.7	23.2	25.1	24.4	21.3	16.8	10.1	3.8
	5%	WB	-4.6	-1.8	3.9	8.2	10.8	13.4	15.2	15.1	12.2	8.0	2.4	-3.1
		MCDB	-0.4	2.9	11.7	18.5	20.6	22.2	24.1	23.4	20.0	15.0	8.3	1.2
	10%	WB	-6.1	-2.8	3.0	7.3	10.2	12.8	14.5	14.4	11.5	7.0	1.4	-4.3
		MCDB	-2.6	1.2	9.9	16.6	19.3	21.5	23.2	22.5	18.9	13.3	6.5	-0.6
Mean Daily Temperature Range	5% DB	MDBR	9.6	9.2	9.6	10.7	10.8	10.7	11.1	10.8	10.7	10.5	9.0	9.1
		MCDBR	11.7	10.8	12.4	13.6	13.1	13.0	13.4	13.0	12.7	12.2	11.6	11.3
		MCWBR	7.5	6.3	5.9	5.5	4.7	4.1	4.1	4.2	4.4	5.0	5.8	6.9
	5% WB	MCDBR	11.4	10.1	11.1	12.5	11.5	11.0	11.6	11.2	11.4	11.1	10.7	11.0
		MCWBR	7.4	6.0	5.4	5.4	4.3	4.0	4.2	4.2	4.5	4.7	5.6	6.7
Clear-Sky Solar Irradiance	taub	0.296	0.347	0.456	0.530	0.543	0.533	0.495	0.535	0.507	0.416	0.339	0.290	
	taud	2.227	2.075	1.794	1.662	1.691	1.762	1.884	1.773	1.781	1.937	2.121	2.258	
	Ebn at Noon	848	843	776	749	753	761	786	737	720	747	774	822	
	Edh at Noon	95	130	195	239	237	222	194	211	196	148	105	85	
All-Sky Solar Radiation	RadAvg	2.39	3.19	4.29	5.39	6.13	6.32	6.31	5.68	4.77	3.75	2.64	2.10	
	RadStd	0.21	0.27	0.33	0.34	0.39	0.36	0.36	0.30	0.31	0.22	0.18	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	-0.48	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	-0.60	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page